Kuppampati Sai Chandraneel

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.72
Narayana college	Class XII (Board of intermediate Education, Andhra Pradesh)	2023	94.00%
Narayana school	Class X (Board of Secondary Education ,Andhar Pradesh)	2021	98.66%

PROJECTS

Doctor Appointment Scheduler

Multi-Agent AI, MCP, Gemini API, Google Calendar, Streamlit

- Architected an **AI-powered multi-agent system** to automate the end-to-end doctor appointment workflow, including doctor discovery, specialization/city filtering, reviews, location analysis, and availability evaluation.
- Implemented **MCP-based tool integrations** enabling agents to retrieve doctor/clinic information, evaluate patient feedback and travel convenience, and coordinate **appointment scheduling**.
- Integrated **Google Calendar** for automated appointment booking and developed a conversational interface unifying **doctor selection, time-slot discovery, and scheduling** into a single user experience.

K-Means Based Customer Segmentation and Recommendation System

- Built a customer-level dataset from **541K retail transactions** using RFM and cancellation features for customer segmentation.
- Removed **~5% outlier customers** using Isolation Forest, followed by feature scaling and Principal Component Analysis for clustering.
- Segmented customers using **K-Means (k=3)**, optimized through elbow and silhouette analysis (**silhouette score: 0.236**).
- Developed a cluster-based **product recommendation engine** returning each customer's **top-3 best-selling unpurchased products**.

Multi-Tenant Team Management & Dynamic RBAC Platform

RBAC, JIT Privilege Elevation, WebSockets, Audit Logging, Security

- Engineered a **multi-tenant team management platform** with workspace-level data isolation and **39+ granular permissions** for users, teams, tasks, and security governance.
- Implemented **contextual RBAC and Just-In-Time (JIT) privilege elevation** with structured, tamper-evident audit logs capturing actors, actions, target entities, and timestamps.
- Developed **real-time workspace collaboration** with multi-channel chat, threaded messaging, unread indicators, and interactive Kanban task boards synchronized through WebSockets.
- Built a secure **email invitation system** using SHA-256 token hashing and **1-hour token expiration**, with sanitized Markdown rendering via DOMPurify.

Radiomics MRI Analysis Web Application

Python, SimpleITK, PyRadiomics, Scikit-learn, Flask/React, DICOM/NIfTI, Multithreading

- Built an end-to-end **MRI preprocessing pipeline** incorporating denoising, N4 bias-field correction, normalization, coregistration, and skull-stripping, with a **DICOM-to-NIfTI conversion module** for standardized scan analysis.
- Implemented **multi-modal image coregistration** using SimpleITK with **Mattes Mutual Information** and gradient descent optimization, alongside Non-Local Means denoising with adaptive noise estimation.
- Integrated **multithreading** to offload compute-intensive image processing tasks, maintaining a responsive and **non-blocking web interface** during scan processing.

SKILLS

Languages: Python, C, C++, JavaScript, HTML, CSS

Libraries & Tools: NumPy, Pandas, Matplotlib, Git, LaTeX, VS Code, Jupyter Notebook, Solid Edge

Databases: MySQL, PostgreSQL, MongoDB

Full Stack: React.js, Node.js, Express.js, REST APIs, JWT, WebSockets

RELEVANT COURSES

Computer Science and Programming: Introduction to Programming(C and data Structures) , Fundamentals of Scientific Computing(Numpy, Matplotlib and Pandas) , Computational Methods in Material Science (Python Programming)

Mathematics: Calculus I, Calculus II, Differential Equations, Elementary Linear Algebra

POSITIONS OF RESPONSIBILITY

Public Relations & Networking Head - Media and PR Council, IIT Hyderabad

Led the management of IIT Hyderabad's official social media handles, ensuring consistent branding and outreach. Developed and executed marketing strategies that enhanced the institute's online visibility and engagement.

Events Coordinator - Milan, General Championship

Handle on-ground operations and oversee end-to-end event execution. Coordinate with multiple domains to ensure seamless planning and delivery. Manage logistics and resolve issues to guarantee smooth event flow.

EXTRACURRICULAR

Inter IIT Culturals 8.0 | IIT Kanpur

- Represented IIT Hyderabad at Inter IIT Culturals 8.0 at IIT Kanpur, competing in various competitions.

Hockey & Volleyball Team Member | IIT Hyderabad

- Active member of institute teams; won Silver at DIESTA, Silver in the Institute League, and Bronze at MILAN tournaments.